Dev Soni

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M.Sc. Statistics student (CGPA 8.36, University of Delhi) with hands-on expertise in Python, MySQL, and end-to-end ML pipelines across finance, healthcare, and operations. Adept at translating statistical models into actionable business insights and deploying live applications. Seeking a data analyst / data science internship where statistical rigour drives measurable decisions.

Academic Qualifications

Passing Year	Course	Institute	Percentage/CGPA
2027	M.Sc. Statistics	Ramjas College, University of Delhi	8.36
2025	B.Sc. (Hons) Statistics	Institute of Science, Banaras Hindu University	7.95
2022	Class XII, UP Board	Government Inter College, Sultanpur	73.8%
2020	Class X, UP Board	Government Inter College, Sultanpur	79%

Core Competencies

Time Series Forecasting | ML Pipelines | Statistical Inference & Hypothesis Testing | Feature Engineering | EDA & Data Cleaning | Regression & Classification | Model Evaluation (ROC-AUC, CV) | SQL & ETL Design | Data Visualisation | Streamlit Deployment | Predictive Modelling | Business Insight Communication

Academic & Industry Projects

Google Stock Price Forecasting | SARIMAX · Random Forest · XGBoost · Python — Feb – Mar 2026

Engineered a signal-generation system to automate directional entry-point detection from 2 years of OHLCV data.

- Built an end-to-end **time series forecasting pipeline** using Google (GOOGL) stock data, covering EDA → stationarity testing → feature engineering → statistical & ML modelling.
- Conducted EDA on 2 years of daily OHLCV data (**500+ trading sessions**); performed **Augmented Dickey-Fuller (ADF) test** to identify stationarity and determine optimal differencing (d = 1).
- Applied **Granger causality** analysis to select significant exogenous variables and engineered optimal lag features (**High_lag6, Low_lag9, Open_lag7**).
- Designed and executed **parallelised SARIMAX grid search**, identifying optimal model: **SARIMAX(0,1,0)(2,1,0,20)**; achieved **RMSE = 22.87, MAPE = 6.70%**.
- Diagnosed ML model failure (**$R^2 = -10.13$**) due to **regime shift (train–test distribution mismatch)**, demonstrating strong debugging ability.
- Recovered performance by **engineering return-based features** (percentage change) to handle non-stationarity, including rolling volatility, intraday & gap returns, and multi-period momentum; boosted to **$R^2 = 0.84+$**.
- Implemented **TimeSeriesSplit cross-validation** to prevent data leakage; trained and tuned Random Forest, Gradient Boosting, XGBoost, Voting Regressor; best: **Random Forest Test $R^2 = 84.89\%$**.
- Key insight: **feature engineering (returns) > model tuning** in non-stationary financial time series.

⇢ github.com/Dev-2004-DA/google-stock-forecasting | Live App: t-i-m-e-series-forecasting.streamlit.app

Cardiovascular Disease Prediction | Scikit-learn · Random Forest · ROC-AUC · Python — Jan – Feb 2026

Developed an early-warning classifier across 70,000 patient records to support clinical screening decisions.

- Performed domain-driven data cleaning on **70,000+ patient records**, correcting physiological inconsistencies in blood pressure and BMI values using clinical constraints.
- Applied stratified train–test split and **5-fold cross-validation** to ensure unbiased model evaluation.
- Benchmarked **10+ classifiers**; selected Random Forest achieving **~75–76% CV accuracy, OOB ~73%**. Identified overfitting via train vs. CV gaps; diagnosed bias–variance trade-offs across models.

- Feature importance analysis revealed **systolic/diastolic BP as dominant predictors (~70% combined)**; documented SVM scalability constraints influencing model selection.
- Performed ROC analysis and threshold tuning to boost **disease recall from 69% → 85%** for screening-oriented deployment; analysed precision–recall trade-offs to align with clinical risk tolerance.

⇢ github.com/Dev-2004-DA/cardiovascular-disease-prediction | Live App: cardiovasculardiseasesprediction.streamlit.app

Medical Insurance Cost Prediction | Scikit-learn · ElasticNet · Random Forest · Python Jan – Feb 2026

Built end-to-end regression ML pipelines on 1,338 patient records for insurance pricing support.

- Performed extensive EDA, feature engineering, outlier analysis, and categorical encoding using **Pipelines and ColumnTransformers**.
- Discovered that the target variable (***charges***) is **bimodal (Smokers vs. Non-Smokers)**, explaining non-linearity and residual non-normality in linear models.
- Applied **GridSearchCV + 5-fold CV** to tune hyperparameters across Linear Regression, ElasticNet, Polynomial, Decision Tree, and Random Forest models.
- Achieved **~86% test R^2** with Random Forest; minimal train–test gap proving robustness on high-variance data.
- Conducted residual diagnostics and feature importance analysis; established **smoking status as primary cost driver (~70% feature importance)**, supporting data-driven pricing decisions.

⇢ github.com/Dev-2004-DA/medical-insurance-cost-prediction

House Price Prediction | Scikit-learn · Ridge/Lasso · PCA · Python Jan – Feb 2026

Built regression ML pipelines on a 21K-row housing dataset.

- Applied feature engineering, **regularisation (Ridge/Lasso)**, Polynomial features, and **PCA** to manage multicollinearity and bias–variance tradeoff.
- Evaluated models using train–test split, **5-fold cross-validation**, and residual diagnostics; achieved **~69% CV R^2 (±1.3%)**, demonstrating stable generalisation beyond the linear baseline.

⇢ github.com/Dev-2004-DA/ML_beginning

Airbnb NYC Data Cleaning Pipeline | MySQL · Window Functions · ETL Sep – Oct 2025

Designed a production-grade SQL ETL pipeline for large-scale data standardisation.

- Designed a **3-stage ETL pipeline (raw → staging → clean)** on **100K+ listings**; applied ROW_NUMBER, COALESCE, TRIM, STR_TO_DATE for deduplication & standardisation across multiple tables.
- Applied advanced SQL Window & String Functions for missing value imputation and data quality improvement.

⇢ github.com/Dev-2004-DA/airbnb-data-cleaning-sql

Café Sales Data Cleaning | MySQL · Window Functions · ETL Oct – Nov 2025

Cleaned and standardised a transactional dataset for operational analysis.

- Cleaned a café sales dataset of **10K+ rows** using MySQL, handling duplicates, missing values, and inconsistent entries.
- Standardised columns, enforced constraints, and corrected data types; used SQL window functions (LAG, COALESCE) to impute missing values.

⇢ github.com/Dev-2004-DA/cafe-sales-data-cleaning-sql

Synthetic Customer Profiling | Python · Pandas · Matplotlib · Seaborn Jun 2025

Exploratory data analysis and customer segmentation on synthetic retail data.

- Used Pandas, NumPy for data cleaning & analysis; Matplotlib and Seaborn for comprehensive visualisation.

⇢ github.com/Dev-2004-DA

Tools & Skills

Languages & Tools: Python (NumPy, Pandas, Matplotlib, Seaborn, Scipy, Statsmodels, Scikit-learn, TensorFlow, Keras), MySQL, MS Excel (Pivot Tables, Power Query, XLOOKUP, Basic Dashboards), Power BI, Git/GitHub, Streamlit

Machine Learning: Supervised Learning (Linear/Logistic Regression, Random Forest, Gradient Boosting, XGBoost, SVM), Ridge/Lasso/ElasticNet, Model Evaluation (CV, ROC-AUC, Precision-Recall), Feature Engineering, ANN, Bias-Variance Tradeoff

Statistics & Forecasting: Time Series (ARIMAX, SARIMAX, VARMAX), Hypothesis Testing, ADF Test, Granger Causality, Residual Diagnostics, Statistical Inference, Sampling Theory

Deep Learning: ANN (TensorFlow, Keras), Bias-Variance Tradeoff, Model Evaluation

Educational Background

M.Sc. Statistics, Ramjas College, University of Delhi | CGPA: 8.36 2025 – 2027 (ongoing)
Probability Theory · Time Series · Statistical Inference · Design of Experiments · Stochastic Processes · Non-Parametric Inference · Statistical Computing (R & Python)

B.Sc. (Hons) Statistics, Institute of Science, Banaras Hindu University, Varanasi | CGPA: 7.95 2022 – 2025
Statistical Methods · Statistical Inference · Sample Surveys · Design of Experiments · Decision Theory · Operations Research · Programming with C

Class XII, UP Board, Government Inter College, Sultanpur 2022
Percentage: 73.8%

Class X, UP Board, Government Inter College, Sultanpur 2020
Percentage: 79%

Certifications & Activities

- **Deloitte Australia Data Analytics Job Simulation** · Forage, Aug 2025 — simulated end-to-end client data analysis & dashboard reporting engagement.
- Presented research on **Stochastic Modelling of Reliance Industries stock prices** at academic seminar, Ramjas College — 5 models from Random Walk to GBM.
- **Activities:** Badminton · Swimming